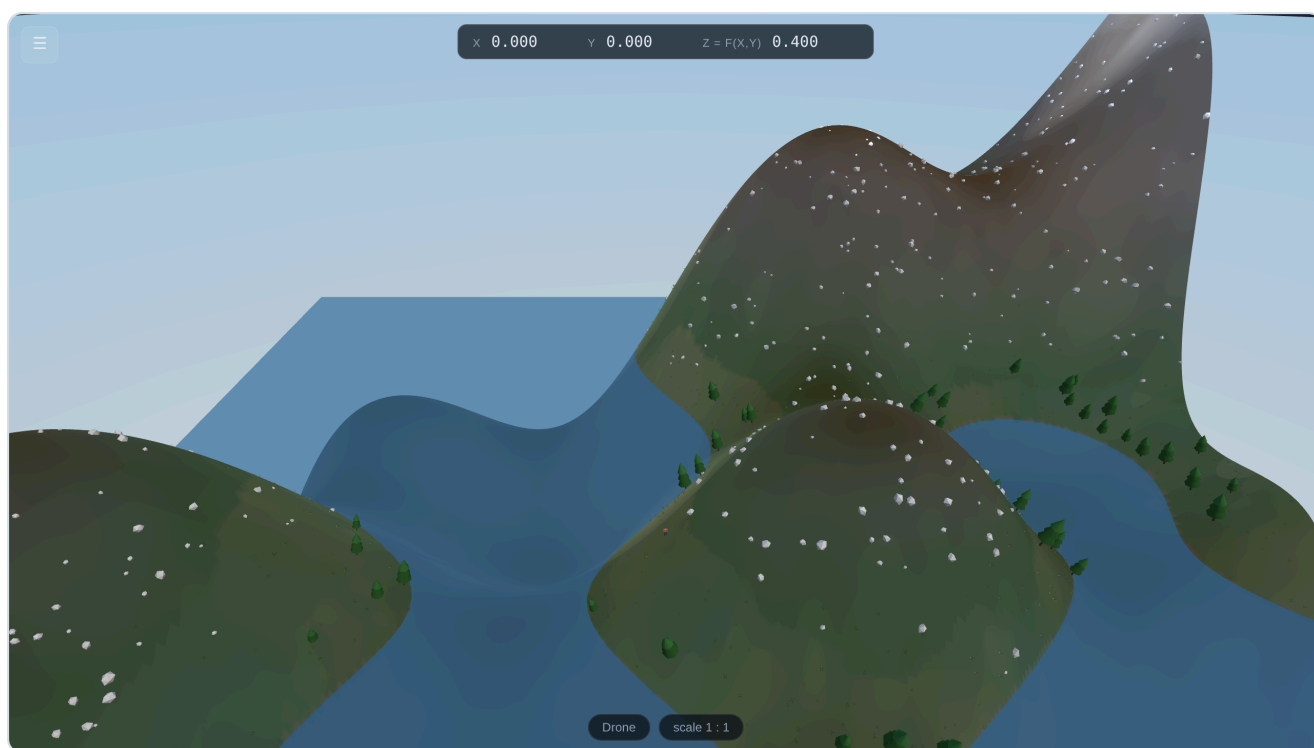


Gradient Peaks

A 3D sandbox for surface plots of functions of two variables — walk on the graph of $f(x, y)$, read its partial derivatives off the ground, and find the maximum over a feasible set.



Written for someone who has never installed or run a program like this before.

Every step is spelled out. Nothing here requires you to write code.

Version 1.0 · Runs on Windows, macOS, Linux, iPad, iPhone and Android · No installation required

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Appendix A — Keyboard reference card (print this one page for the lab)

Appendix B — Every file you received, and what it does

1 Start here — the sixty-second version

You have been given a folder called `Gradient-Peaks`. If it arrived as a `.zip` file, you must **unzip it first** — see the box below. Inside are these items:

ITEM	WHAT IT IS
<code>Gradient-Peaks.html</code>	The program. Double-click it and it opens in your web browser and runs. This one file contains everything. It does not need the internet.
<code>Gradient-Peaks-Guide.pdf</code>	This document.
<code>website/</code>	A folder holding the same program split into its separate parts. You only need this if you want to <i>publish it on the internet</i> for your students (Part 8) or install it as an app (Part 9).
<code>README.txt</code>	A one-page plain-text summary of what you are reading now.

Unzip it first — do not run it from inside the zip

Double-clicking a `.zip` file often shows you the contents without actually extracting them, and programs run from that preview will misbehave.

Windows: right-click the `.zip` → *Extract All...* → *Extract*.

macOS: double-click the `.zip`. A normal folder appears next to it. Use that folder.

Open it

- 1 **Double-click** `Gradient-Peaks.html`.
- 2 Your web browser opens. You will see a dark screen with a small spinning circle and the words *Building the terrain...* for one to three seconds.
- 3 A green and grey mountain appears, seen from the air, with a control panel down the left-hand side. **That is it. It is running.**

There is nothing to install, no account to create, and no internet connection needed. The program is not "downloading" anything — it is drawing the mountain from the formula $f(x,y) = (x*y)^{0.5}$, which you can see typed in the box at the top of the panel.

English and Spanish

The program follows your browser: if that is set to Spanish, it opens in Spanish. You can switch at any time with the **English / Español** selector in the top corner of the control panel, without losing whatever you are in the middle of.

If you publish it to a web address (Part 8), adding `?lang=es` to the end of the link makes it open in Spanish for your students whatever their browser is set to — and `?lang=en` forces English. **A Spanish edition of this guide is included as `Gradient-Peaks-Guia.pdf`.**

If it opens in a browser you do not like

Any modern browser works, but **Google Chrome** and **Microsoft Edge** give the smoothest result. To choose one: right-click `Gradient-Peaks.html` → *Open with* → pick your browser. On macOS the menu item is the same; hold `Control` and click if you have no right mouse button.

One thing that will not work, and why

Inside the `website/` folder there is another file called `index.html` . **Double-clicking that one gives you a blank screen.** That is not a fault. Browsers refuse, for security reasons, to let a page opened directly from your hard drive load its own separate program files. The single `Gradient-Peaks.html` avoids the problem by having everything already inside it.

Use `Gradient-Peaks.html` on your own machine. Use the `website/` folder only when you publish it to a web address (Part 8).

2 Will it run on my computer?

Almost certainly yes. The program needs a graphics feature called **WebGL**, which every browser has had since about 2013.

YOU NEED	DETAILS
A computer or phone	Roughly 2014 or later. It has been kept deliberately small — under one megabyte — so it runs on school laptops and cheap tablets.
A modern browser	Chrome, Edge, Firefox or Safari, updated within the last few years.
Nothing else	No internet, no Java, no Flash, no plug-in, no administrator password.

Checking in ten seconds

Open `Gradient-Peaks.html` . If you see a mountain, you are done — that is the test. If you see a blank or black screen instead, go to Part 10.

A note about school and university computers

Because this is an ordinary web page, it does not trigger the software-installation restrictions that lock down most institutional machines. There is no installer and no `.exe` . If your students can open a web page, they can run this.

3 Your first five minutes

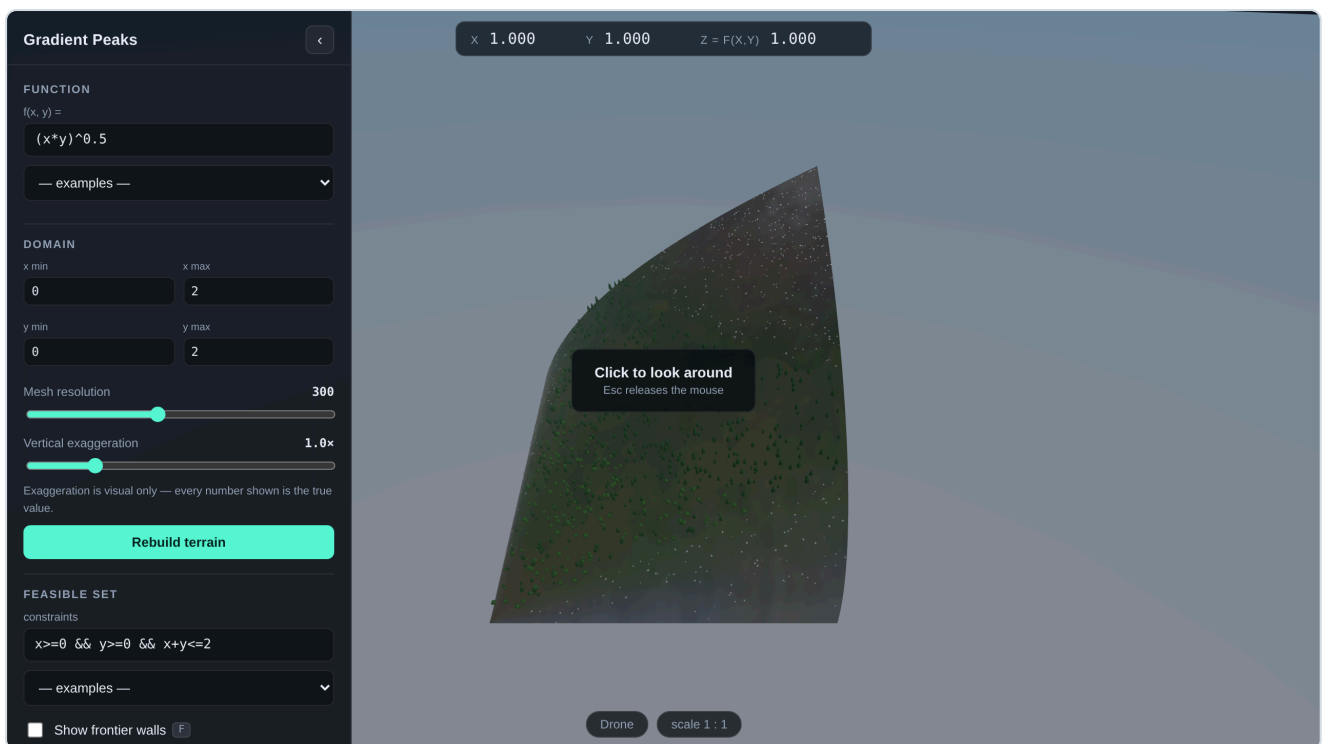


Figure 1. The opening screen. The control panel is on the left. Along the top, the readout always shows where you are: x , y , and the height $z = f(x, y)$. At the bottom, the current view mode and scale. You begin in the air, looking at the whole surface.

Step one: look at the whole surface

You start as a **drone**, hovering above the plot. This first view is deliberate: before you go down and stand on the surface, you should see it as what it is — the graph of a function.

Step two: take control of the mouse

Click once anywhere on the mountain. The message "*Click to look around*" disappears and your mouse pointer vanishes. This is normal and is exactly how a 3D game behaves: your mouse now turns your head instead of moving a pointer.

Move the mouse to look around. Press **W A S D** to fly. Press **Space** to rise and **Ctrl** to descend. Hold **Shift** to go faster.

The single most important key

Esc gives you your mouse pointer back. If you ever feel trapped, press **Esc**. Nothing is lost. Click the scene again to resume looking around.

Step three: go and stand on it

Press **2**. You drop onto the surface behind a small orange-jacketed explorer, who is exactly **1.80 metres tall**. That figure is the ruler for everything else in the scene: the trees, the boulders, and the measuring circle you will meet in a moment are all sized against a person.

Walk with **W** **A** **S** **D**. Watch the numbers along the top change as you move. **That readout is the whole point of walking:** the student feels the slope in their hands while watching **z** climb.

Press **1** to see through the explorer's own eyes, **3** to go back to the drone, and **2** to return to the over-the-shoulder view.

Step four: get the panel back

Press **Esc** to release the mouse, then press **Tab** to hide or show the control panel. You can also click the **<** button in its top corner, or the **≡** button that replaces it.

Keyboard shortcuts pause while you type

When your cursor is inside a text box, the letter keys type letters — they do not trigger shortcuts. So typing **x*y** will not accidentally toggle three settings. Press **Enter** when you have finished typing to apply what you wrote and leave the box.

4 The control panel, one control at a time

Language

The selector beside the panel title switches the whole interface between **English** and **Español** — including the error messages the parser produces. Nothing is recomputed, so you can switch mid-demonstration.

Function

$f(x, y) =$ — the formula that becomes the landscape. Type a new one and press **Enter**. Part 5 explains the notation in full.

— **examples** — — a menu of ready-made functions. Choosing one loads it and sets a sensible domain. This is the fastest way to explore.

If what you type does not make sense, a red message appears underneath naming the problem and the character where it went wrong. **The mountain you already had stays on screen** — a typing mistake never destroys your work.

Domain

x min, x max, y min, y max — the rectangle of the plane you are plotting over. The default is the square from 0 to 2 in both directions.

Mesh resolution — how finely the surface is divided into triangles. Higher is smoother but slower to build. 300 is the default; lower it to 100 or 80 on an old computer.

Vertical exaggeration — stretches the landscape upward for looks. **Leave this at 1.0 for teaching**. At any other value the picture is stretched but the numbers on screen remain the true, unstretched ones, which is exactly the sort of mismatch that confuses students. It exists for functions so flat you cannot see them at all.

Rebuild terrain — applies the function and domain. Pressing **Enter** in any of the boxes does the same thing.

Feasible set

constraints — the inequalities that define where the student is allowed to look for the maximum. Part 6 covers the notation.

Show frontier walls **F** — draws glowing walls along the border of the feasible region, standing on the surface. This is the "frontier of the country".

Make the outside translucent **G** — everything outside the feasible set fades to a ghost, so only the permitted region is solid. Turning both of these on is the clearest way to pose a constrained optimisation problem.

Map & terrain

Level curves **C** — draws the contour lines on the surface, at round-numbered heights. The spacing chosen (for example $\Delta z = 0.2$) is shown next to the checkbox. You can walk along them.

Topographic colours **M** — repaints the surface in the standard hypsometric palette of a paper map: blues below zero, then green, yellow, tan, brown and white with height. Combine with **T** (look straight down) to turn the 3D

scene into the flat map students already know.

Vegetation, rocks & snow — turns the scenery on and off. The scenery sits *on top of* the surface and never changes its shape.

Water in the depressions — a translucent lake surface at height zero, so any region where $f < 0$ fills with water.

Decoration density — how much scenery. Turn it down to speed things up.

Shadows — prettier, noticeably slower. Off by default.

Derivatives

Highlight neighbourhood H — lights up a disc on the surface centred on the explorer. This is the neighbourhood in which all the rates of change are measured.

Radius — 1 metre or 2 metres, measured against the 1.80 m explorer.

■ **$\partial f / \partial x$ arrow** X — a blue arrow drawn *along the surface* in the direction of increasing x .

■ **$\partial f / \partial y$ arrow** Y — the same in red, for increasing y .

■ **Gradient ∇f** V — in teal and at double width, pointing in the steepest uphill direction.

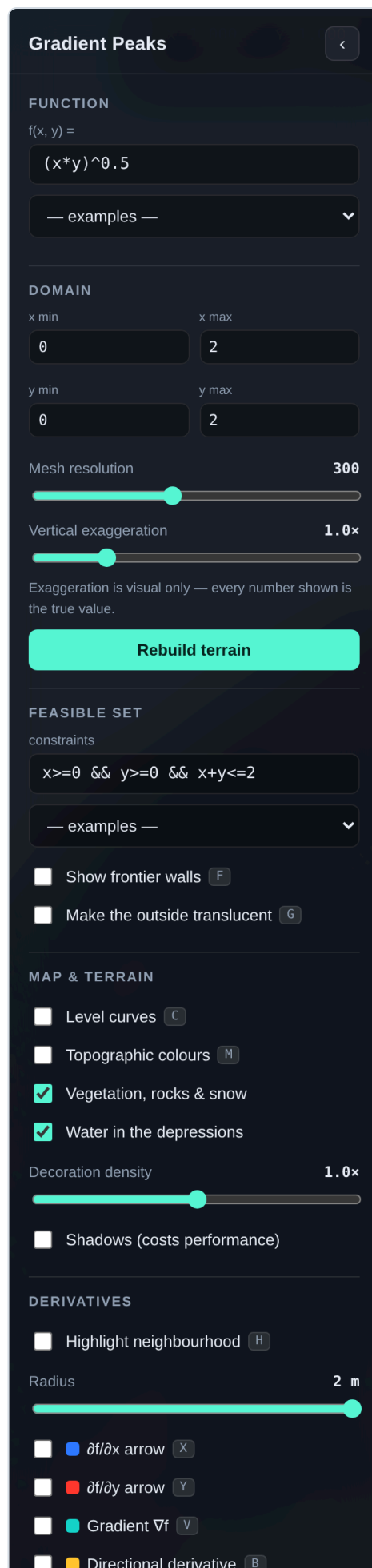
Each one adds a badge to the top of the screen showing two numbers: the **instantaneous** derivative at the explorer's feet, and the **average** rate of change over the radius of the disc. Watching those two numbers converge as you shrink the radius is a lesson in itself.

■ **Directional derivative** B — freezes the explorer and lets you swing a direction vector $\mathbf{u}$ around the rim with the mouse. The badge shows $D_{\mathbf{u}}f$ and the angle. To look around while in this mode, hold the **right** mouse button.

Tangent plane P — draws the flat plane that best matches the surface at the explorer's feet.

Zoom-in ruler

Shrinks the explorer by a factor of ten per notch: 1.8 m, then 18 cm, 1.8 cm, 1.8 mm, and finally 0.18 mm. The camera, the walking speed and the measuring disc all shrink with them, so from the explorer's point of view nothing changes — but the patch of surface they occupy gets a hundred, a thousand, ten thousand times smaller. Activity 8 in Part 7 explains why this matters.



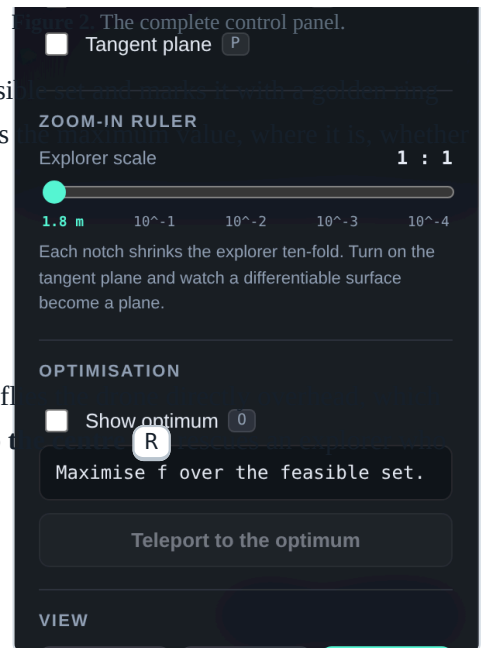
Optimisation

Show optimum 0 — finds the highest point of the surface within the feasible set and a shaft of light you can see from anywhere. The report underneath gives you whether it sits in the *interior* or on the *boundary*, and the size of the gradient there.

Teleport to the optimum — puts the explorer on that spot instantly.

View

First person 1, **Third person** 2, **Drone** 3. **Look straight down** T flattens the surface, projects the surface onto the plane $z = 0$ — the flat map view. **Return to terrain** R has wandered off.



5 Writing your own functions

Type into the `f(x, y) =` box and press `Enter`. The notation is the ordinary one you would use in a calculator or a spreadsheet, with two conveniences added.

The rules

SYMBOL	MEANING	EXAMPLE
<code>+ - * /</code>	add, subtract, multiply, divide	<code>x/2 - y</code>
<code>^</code>	raise to a power	<code>x^2</code> , <code>x^0.5</code> , <code>x^-1</code>
<code>()</code>	grouping	<code>(x+y)^2</code>
<code> </code>	absolute value	<code> x-y </code>
<code>pi</code> , <code>e</code>	the usual constants	<code>sin(pi*x)</code>

Two conveniences

You may leave out the multiplication sign. `2x` means `2*x`, `x y` means `x*y`, and `3(x+y)` means `3*(x+y)`.

Powers bind tightest. `-x^2` is `-(x^2)`, not `(-x)^2` — the same convention as in your lecture notes.

Functions you can use

`sin` `cos` `tan` `asin` `acos` `atan` `atan2` `sinh` `cosh` `tanh` `exp` `ln` `log` `log2` `log10` `sqrt` `cbrt`
`abs` `sign` `floor` `ceil` `round` `min` `max` `pow` `hypot` `mod` `clamp` `step` `gauss`

`hypot(a,b)` is `sqrt(a^2+b^2)`. `gauss(t)` is the bell curve `e^(-t^2/2)`; `gauss(t, centre, width)` shifts and stretches it.

Examples that work

TYPE THIS	YOU GET
<code>(x*y)^0.5</code>	The Cobb–Douglas surface. Undefined unless <code>x</code> and <code>y</code> are both positive or both negative.
<code>x^0.3*y^0.7</code>	Cobb–Douglas with unequal shares.
<code>1-x^2-y^2</code>	A dome. One clean interior maximum at the origin.
<code>x^2-y^2</code>	A saddle. Useful for showing a point where the gradient vanishes but there is no maximum.
<code>sin(3x)*cos(3y)/2</code>	An egg carton: many local maxima.
<code>0.6*sin(2x)+0.4*cos(3y)+0.3*x*y</code>	Rolling hills with lakes. The prettiest one to walk in.
<code>-(x^2+y^2)^0.5+1</code>	A cone. The tip is a maximum where the function is <i>not</i> differentiable — see Activity 8.
<code>x*y*exp(-(x^2+y^2))</code>	Four alternating peaks and troughs.

Mistakes, and what the program tells you

YOU TYPE	YOU SEE	THE FIX
<code>sin(x</code>	Missing ")" after sin((at character 6)	Close the bracket.
<code>x*yy</code>	Unknown name "yy" (at character 3)	Only <code>x</code> and <code>y</code> are variables.
<code>2+</code>	Unexpected end of expression	Something is missing after the <code>+</code> .
<code>sin</code>	"sin" is a function — write sin(...)	Give it an argument.

Where a function is undefined, there is simply no ground

$(x*y)^{0.5}$ has no real value when x and y have opposite signs. Plot it over a domain that crosses the axes and two quadrants come out **empty** — and the explorer will not walk into them. A note appears at the bottom of the screen telling you what fraction of the domain is undefined. This is not a bug to be worked around; it is a genuinely useful picture of a function's domain.

6 Writing feasible sets

The **constraints** box takes inequalities in x and y . Whatever satisfies them is the feasible set — the region in which the maximum must be found.

SYMBOL	MEANING
$\leq$ $\geq$	less than or equal, greater than or equal
$<$ $>$	strictly less, strictly greater
$\&\&$	and — both conditions must hold
$ $	or — either will do
$!$	not

You may chain comparisons the way you would write them by hand: $0 \leq x \leq 1$ means exactly what it looks like. Leaving the box empty means "no constraint at all".

Examples

TYPE THIS	YOU GET
$x \geq 0 \ \&\& \ y \geq 0 \ \&\& \ x + y \leq 2$	A triangle — the classic budget set.
$x^2 + y^2 \leq 1$	A disc of radius 1.
$x + y \leq 2 \ \&\& \ 2x + y \leq 3 \ \&\& \ x \geq 0 \ \&\& \ y \geq 0$	Two budget lines at once: a four-sided region.
$x * y \geq 0.3 \ \&\& \ x + y \leq 2.5$	A curved constraint against a straight one.
$0 \leq x \leq 1.5 \ \&\& \ 0 \leq y \leq 1.5$	A rectangle, written with chained comparisons.

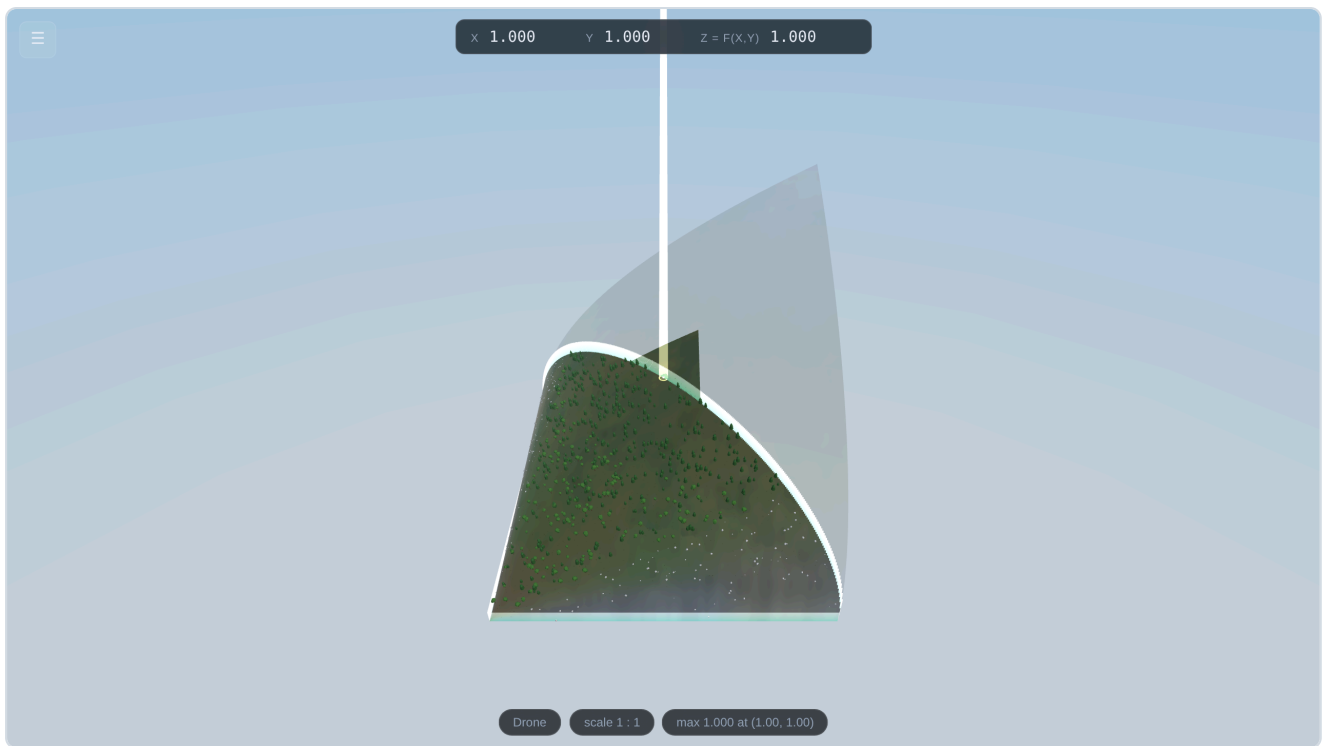


Figure 3. $x \geq 0 \ \&\& \ y \geq 0 \ \&\& \ x + y \leq 2$ with both feasible-set switches on. The permitted triangle is solid ground with its scenery; everything outside has faded to a ghost. The shaft of light marks the constrained maximum.

7 Nine classroom activities

One for each feature. Each takes five to ten minutes and ends with something the students can state in words.

Activity 1 — A function is a landscape

Set up: the program as it opens.

Do: Look at the surface from the drone. Press **2** and walk uphill. Ask the class to call out what **z** is doing as you climb, and then to say what **z** is.

The point: the height of the ground under your feet is the value of the function at the point you are standing on. Everything else in the course is a statement about this landscape.

Activity 2 — Smooth surfaces, rough scenery

Do: From the air, note that the mountain looks perfectly smooth. Now turn **Vegetation, rocks & snow** off and on.

The point: The trees and boulders are placed *on* the surface; they never change its shape. A differentiable function has no corners no matter how close you look — but real mountains are covered in things that do. Being able to separate the two is worth a minute of discussion.

Activity 3 — Where you are allowed to stand

Set up: constraints `x>=0 && y>=0 && x+y<=2`. Turn on **Show frontier walls** and **Make the outside translucent**.

Do: Walk to the wall and try to see over it. Ask: where on this triangle is the ground highest?

The point: A constrained optimisation problem is a question about a region, not about a formula. Students who can see the region rarely get the geometry wrong afterwards.

Activity 4 — The same surface from three heights

Do: Switch between **1**, **2** and **3**, then press **T** to look straight down.

The point: The top-down view is the flat picture in the textbook. Showing students that it is the *same object* seen from above dissolves a surprising amount of confusion about contour diagrams.

Activity 5 — Level curves are paths you can walk

Set up: turn on **Level curves** **C** and **Topographic colours** **M**.

Do: Stand on a contour line and walk *along* it, watching **z** in the readout. Then walk across the lines and watch it change quickly.

The point: A level curve is where the function does not change. Walking one and watching the number hold still makes that concrete. Press **T** afterwards and the same lines become an ordinary contour map.

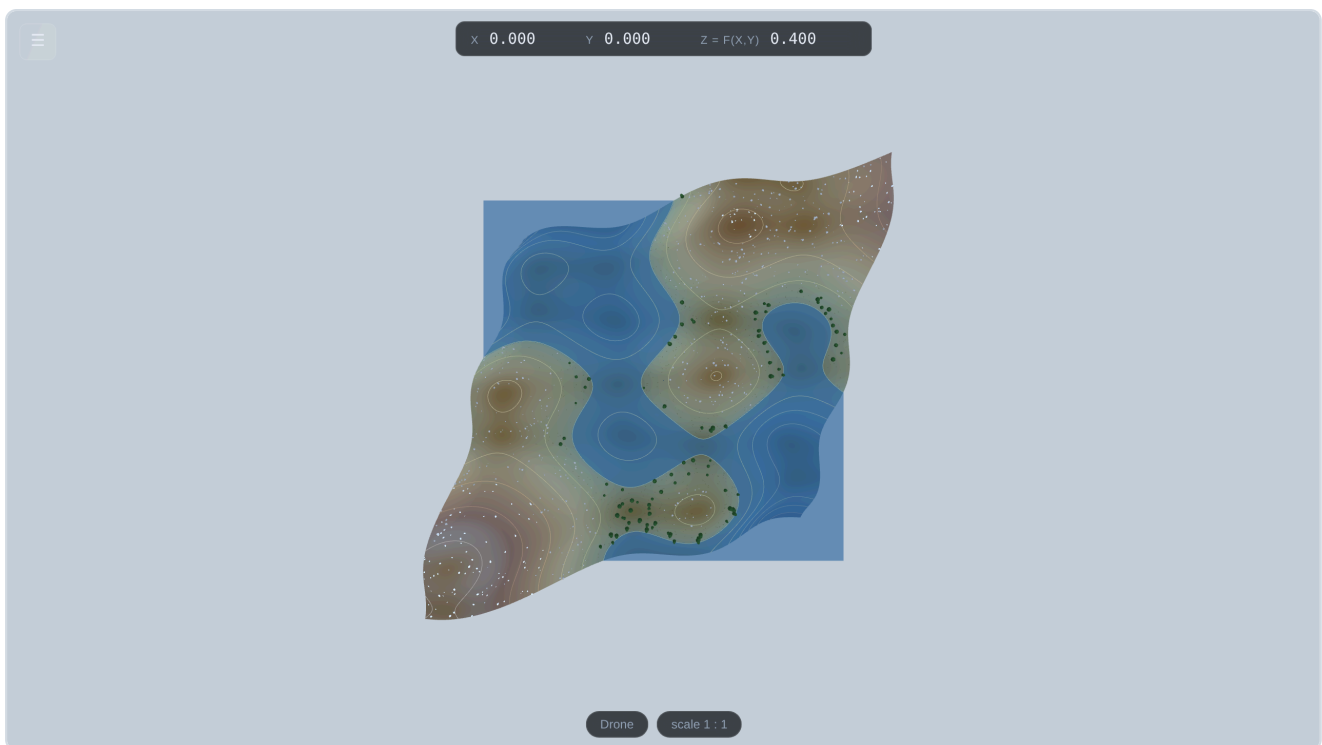


Figure 4. The same landscape from directly overhead with contour lines and map colours on. This is the picture in the textbook — and the students have just been standing in it.

Activity 6 — Reading the partial derivatives off the ground

Set up: $f(x, y) = (x \cdot y)^{0.5}$. Turn on $\partial f / \partial x$ (X), $\partial f / \partial y$ (Y) and **Gradient** (V). Walk to roughly (1, 1).

Do: Read the three badges. At (1, 1) they say $\partial f / \partial x = 0.500$, $\partial f / \partial y = 0.500$, $\|\nabla f\| = 0.707$. Check by hand: the partial derivatives of $\sqrt{xy}$ at (1, 1) are both $\frac{1}{2}$, and $\sqrt{(\frac{1}{4} + \frac{1}{4})} = 0.707$.

Then: point out that the teal gradient arrow sits exactly between the blue and red ones, and that it is the steepest way up. Walk to a different point and watch all three swing round.

The point: the gradient is not an algebraic accident, it is a direction you can see and walk in.



Figure 5. The three arrows, drawn following the curve of the ground rather than as straight lines through the air. Blue is $\partial f/\partial x$, red is $\partial f/\partial y$, and the double-width teal arrow is the gradient. Here the red arrow has almost no length, because $\partial f/\partial y$ is nearly zero — so the gradient points along the blue one.

Activity 7 — Every other direction

Set up: turn on **Directional derivative** **B**.

Do: The explorer freezes. Swing the mouse slowly and watch $D_{\mathbf{u}}f$ in the badge. Find the direction where it is largest — it lands on the gradient. Find where it is zero.

The point: The direction of zero directional derivative is the level curve, at right angles to the gradient. Students usually learn that as a formula; here they find it by hand in about fifteen seconds.

Activity 8 — Why differentiable means "flat if you look closely"

Set up: turn on **Tangent plane** **P**. Leave the zoom ruler at 1:1.

Do: Note that the yellow plane matches the surface underfoot but peels away from it a few metres out. Now drag the **Zoom-in ruler** one notch at a time and watch what happens.

The point: by 1:1000 the surface and its tangent plane are indistinguishable. That is differentiability: not a formula, but the fact that a small enough piece of the surface is a plane. Follow it with the cone $-(x^2+y^2)^{0.5+1}$ and zoom in on the tip — it stays a point however far you go. That is what a non-differentiable point looks like.



Figure 6a. At 1:1 the tangent plane (yellow) clearly leaves the surface.



Figure 6b. The same spot with the explorer shrunk to 1.8 mm. The surface has become its tangent plane.

Activity 9 — The maximum, and why the gradient is not zero there

Set up: $f(x,y) = (x*y)^{0.5}$, constraints $x \geq 0$ && $y \geq 0$ && $x+y \leq 2$, **Show frontier walls on**, **Show optimum** on.

Do: Read the report. It says the maximum is 1.0000 at (1.0000, 1.0000), that it lies **on the boundary of the feasible set**, and that $\|\nabla f\|$ there is 0.7071 — *not* zero. Teleport there, turn on the gradient arrow, and see it pointing straight at the wall.

Then: switch the feasible set off and press again. The maximum jumps to the far corner of the domain.

The point — and it is the reason this program exists: at an unconstrained maximum the ground goes flat and the gradient dies. At a *constrained* maximum it need not: the ground is still climbing, and what stops you is the border. Students who have stood at that spot and seen the gradient arrow jammed against the wall do not later write "set the gradient to zero" on a constrained problem.

8 Giving it to your students

Four ways, easiest first. Option A needs nothing at all; option D gives you a permanent web address. All four are free.

Option A — Send them the file

Email `Gradient-Peaks.html` as an attachment, or drop it in your course folder, or put it on a USB stick. Students double-click it. That is the whole procedure.

One caution about email

Some mail systems strip or block `.html` attachments. If it does not arrive, put the file in a shared drive (OneDrive, Google Drive, Dropbox) and send the link instead — but tell students to **download** it and then open it, not to preview it in the browser, since some drive previewers will show the file's text rather than run it.

Option B — Upload it to your course page

Moodle, Canvas, Blackboard and their relatives all accept a file upload. Add `Gradient-Peaks.html` as a resource. Students click it and it runs.

Option C — Netlify Drop: a real web address in about two minutes

This gives you a link you can put on a slide. No account is needed to start.

- 1 Go to `app.netlify.com/drop` in your browser.
- 2 Drag the `website` folder — the whole folder, not the files inside it — onto the dashed box on that page.
- 3 Wait a few seconds. You are given an address like `https://cheerful-lion-12ab34.netlify.app`.
- 4 Open it to check, then share it. Anyone with the link can use it, on any device.

Create a free account within 24 hours if you want to keep the address permanently or give it a friendlier name.

Option D — GitHub Pages: permanent, free, and version-controlled

More steps, but the address never expires and updating it later is easy. If the project already lives in a GitHub repository:

- 1 Open the repository on `github.com`.
- 2 Click **Settings** (the tab along the top of the repository, not your account settings).
- 3 In the left sidebar, click **Pages**.
- 4 Under **Source**, choose **GitHub Actions**.
- 5 The workflow already included in the project (`.github/workflows/pages.yml`) publishes the `app/` folder automatically on every push to the main branch.
- 6 After a minute or two the same Pages screen shows your address, of the form `https://<your-username>.github.io/<repository>/`.

Which folder do I upload?

For options C and D you publish the **website/** folder, because a real web server can serve its separate files correctly. Option A and B use the single `Gradient-Peaks.html` instead. Both contain the same program.

9 Installing it as an app

Once the program is at a web address (options C or D above), it can be installed like an app — its own icon, its own window, no browser bar, and it works offline afterwards. There is no app store involved, no payment, and no developer account.

DEVICE	HOW
iPhone / iPad	Open the address in Safari (this does not work in Chrome on iOS). Tap the Share button — the square with an arrow — then scroll down and tap Add to Home Screen .
Android	Open the address in Chrome . Tap the three-dot menu, then Install app or Add to Home screen .
Windows	Open in Chrome or Edge . Click the small install icon at the right-hand end of the address bar (a monitor with an arrow), then Install .
macOS	In Chrome or Edge, the same install icon. In Safari , choose File → Add to Dock .

After installing, the program keeps a copy of itself on the device. Students can use it on the bus with no signal.

10 When something goes wrong

WHAT YOU SEE	WHY	WHAT TO DO
A blank white or black page, and you opened <code>website/index.html</code>	Browsers will not let a page opened from your hard drive load its own separate files.	Open <code>Gradient-Peaks.html</code> instead. That is what it is for.
Stuck on "Building the terrain..." for more than about 20 seconds	The mesh resolution is too high for this computer, or the function is very slow to evaluate.	Reload the page. Drag Mesh resolution down to 100 before changing anything else.
The mouse pointer disappeared and you cannot click anything	Nothing is wrong — the program has taken the mouse so it can turn your head.	Press <code>Esc</code> .
Typing in the function box toggles settings	Your cursor is not actually inside the box.	Click directly into the box first. While it holds the cursor, letters type normally.
A red message under the function box	The formula could not be read. The message names the character where it failed.	See the mistakes table in Part 5. The previous landscape is still on screen and still usable.
Large parts of the terrain are missing	The function has no real value there — for instance <code>sqrt</code> of a negative number.	Nothing to fix. Change the domain if you want a complete surface; the note at the bottom of the screen tells you how much is undefined.
Everything is jerky	The computer's graphics are struggling.	Part 11, in the order given there.
The explorer will not move	Directional derivative is on. It freezes the explorer on purpose.	Turn it off with <code>B</code> .
The screen is filled with a solid colour	The camera is inside the terrain or hard against a frontier wall.	Press <code>R</code> to return to the centre, or <code>3</code> to fly out as the drone.
"WebGL is not available" or similar	Hardware graphics are switched off in the browser.	In Chrome or Edge: Settings → System → turn on <i>Use graphics acceleration when available</i> , then restart the browser. On very locked-down machines, ask IT.

11 Making it run faster on an old computer

Change these in order and stop as soon as it feels smooth. The first two cost you almost nothing.

- 1 **Turn off Shadows** — if it is on, this is the biggest single saving.
- 2 **Decoration density to 0.3, or to 0** — the scenery is the most expensive part of the picture and none of the mathematics depends on it.
- 3 **Mesh resolution to 150, then 100** — the surface gets slightly more angular close up, but every number stays exactly as accurate.
- 4 **Make the browser window smaller** — cost grows with the number of pixels, so a half-size window is roughly twice as fast.
- 5 **Turn off Level curves** while walking, and back on when you need them.

Accuracy is never what you are trading away

Every setting in this part changes only how the landscape is *drawn*. The derivatives, the gradient and the optimum are computed from the formula itself, so the numbers on screen are identical at the lowest settings and the highest.

12 What is inside, and licensing

For colleagues and for IT departments who want to know what they are being asked to allow.

It is an ordinary web page

No installer, no background service, no administrator rights. It never sends anything anywhere: it has no network code at all beyond loading its own files. Nothing is collected, stored or transmitted.

Nothing typed by a student is executed as code

The formula box is read by a purpose-built mathematical parser. It can produce a number and nothing else — there is no `eval` anywhere in the program. A student cannot type something into the function box that makes the program do anything other than draw a surface.

The mathematics is honest

The horizontal and vertical scales are the same, so the slope you see is the slope the program reports. (The vertical exaggeration slider deliberately breaks that, which is why it is labelled "visual only" and defaults to off.) The optimum is found by scanning the feasible set and then refining the best candidates, and every trial point must satisfy the constraints — which is why a boundary maximum is found correctly.

Licence

The program is released under the MIT licence: use it, change it, and give it to your students freely. It uses one external library, **three.js** (also MIT, © three.js authors), whose licence text travels with the files in `website/vendor/`.

Appendix A — Keyboard reference card

One page. Print it and leave it next to the projector.

KEY	DOES
W A S D	Walk or fly
Shift	Hold to move faster
Space / Ctrl	Drone up / down
mouse	Look around (click the scene first)
Esc	Release the mouse pointer
Tab	Hide or show the control panel
1 / 2 / 3	First person / third person / drone
T	Look straight down (map view)
R	Return to the centre
C	Level curves
M	Topographic colours
F	Frontier walls of the feasible set
G	Make everything outside translucent
H	Highlight the neighbourhood disc
X	 $\partial f / \partial x$ arrow
Y	 $\partial f / \partial y$ arrow
V	 Gradient ∇f
B	 Directional derivative (freezes the explorer)
P	Tangent plane
O	Show the optimum
Enter	In a text box: apply what you typed

On a phone or tablet: drag on the left half of the screen to walk, drag on the right half to look around.

Appendix B — Every file you received

FILE	WHAT IT DOES
<code>Gradient-Peaks.html</code>	The whole program in one file. Double-click to run.
<code>Gradient-Peaks-Guide.pdf</code>	This guide.
<code>README.txt</code> / <code>LEEME.txt</code>	A one-page summary, in English and Spanish.
<code>website/js/il18n.js</code>	The English and Spanish text, and the language switch.
<code>website/index.html</code>	The page structure and the control panel.
<code>website/css/style.css</code>	How the panel and readouts look.
<code>website/js/mathexpr.js</code>	Reads what you type into a mathematical expression.
<code>website/js/field.js</code>	Converts between mathematical coordinates and metres; computes gradients.
<code>website/js/terrain.js</code>	Builds the surface, its colours, the water and the frontier walls.
<code>website/js/decor.js</code>	Scatters the trees, rocks, grass and snow.
<code>website/js/analysis.js</code>	Level curves, derivative arrows, tangent plane, and the optimiser.
<code>website/js/player.js</code>	The explorer, the drone, and the three cameras.
<code>website/js/main.js</code>	Puts the scene together and wires up the controls.
<code>website/vendor/</code>	three.js, the 3D graphics library, with its licence.
<code>website/sw.js</code>	Lets an installed copy work offline.
<code>website/manifest.webmanifest</code>	Tells a phone the name and icon to use when installing.
<code>tools/build-standalone.mjs</code>	Optional. Rebuilds <code>Gradient-Peaks.html</code> if you change the source.

Gradient Peaks — built for teaching the geometry of functions of two variables.
Everything in this guide was checked against the running program.